

LEBRE v0.3.2 Architecture Specification

Principled evolution: statistically valid structural decisions under a computational budget

Canonical expansion: Lifecycle-governed Evidence-Based Resource Evolution

Lineage: LEBRE v0.1 (Track B) → v0.2 branch → v0.3 → v0.3.1 → **v0.3.2**

ARCHITECTURE STATUS

Research specification — NON-canonical

CLASSIFICATION

PRINCIPLED_ONLINE_STRUCTURAL_LEARNING_ARCHITECTURE

NOVELTY READINESS

NOVELTY_CLAIM_READY = NO

M3 BOUNDARY

UNOPENED (single latent state)

1. Executive Summary and Document Guide

LEBRE v0.3.2 is the third generation of the LEBRE family (*Lifecycle-governed Evidence-Based Resource Evolution*). It keeps the family identity: an always-on base pathway, a sparse discrete-delay pathway, one recurrent latent-state pathway, a structural lifecycle with shadow probation and explicit resource governance. What changes is that **every structural decision is now a statistical test with a controlled error rate**, all evidence evolves in stream time, and the latent pathway is the Kalman predictor in innovations form.

Status: NON_CANONICAL research specification. LEBRE v0.1 (Track B organisation) remains the frozen canonical reference (FROZEN_WITH_SCOPE_LIMITS). M3 remains UNOPENED; NOVELTY_CLAIM_READY = NO. Reference implementation: lebre_v032.py (SHA-256 ef0d9634...32efe8).

Internal (I1–I14, N=30, pre-registered): NMSE 0.219 vs 0.309 for v0.2 A0 ($\Delta = -0.0905$, 30/30 seeds); 96.9 FP/step vs 111.2.

Explainability: exact structure recovered at 88.6% of checkpoints vs 75.2% for v0.2.

Validity: 0 false promotions in 400 null streams (Gaussian, t_3 and heteroscedastic noise); exact causality; exact scale invariance.

External: 3rd by mean rank on column-permuted BENCH-01 (mean rank 4.0 vs 13.5 for Track B) ; BENCH-02: 25 tasks, global position 6 — pre-registered decision NOT_JUSTIFIED_EXTERNALLY.

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Conventions

Vectors in bold ($\mathbf{x}_t \in \mathbb{R}^D$), scalars in italics, discrete time by subscript t . "Stream time" is the sample index of the data stream. Costs are floating-point operations per step (FP/step), with the v0.1 taxonomy. Evidence is in *nats*. Evidence labels: ESTABLISHED (literature result), VERIFIED (empirical test in this project), DESIGN (non-derived choice).

2. Architectural Identity, Principles and Classification

2.1 Invariant principles

#	Principle	Mechanism
P1	Every structural decision is a test with controlled error.	promotion: Ville + de la Peña; eviction: CUSUM/ARL
P2	All evidence evolves in stream time.	cadences only add bounded waiting (ARB10 lesson)
P3	All structure is costly and pays rent.	excitation-weighted MDL rent; budget $M_{\max}$
P4	Silence is not evidence.	persistent-excitation gate
P5	Candidates do not perturb the prediction.	shadow probation; initial coefficient = posterior mean
P6	Scale invariance and explainability by construction.	dimensionless features; every atom has a physical meaning

2.2 Is LEBRE v0.3.2 an architecture?

Criteria drawn from four traditions with a published definition of the term: systems engineering (ISO/IEC/IEEE 42010:2022: "fundamental concepts or properties of an entity in its environment and governing principles for the realization and evolution of this entity"), cognitive architectures (fixed structures + learned knowledge; Newell, Laird), learning architectures (Cascade-Correlation, Dyna, Horde) and neural architecture search (topology of the computational graph; Elsken, Metzen & Hutter 2019).

#	Criterion	Source	LEBRE v0.3.2
K1	Distinct elements with interfaces	ISO/IEC/IEEE 42010	✓ 9 elements: base, delays, latent, screening, tests, supervisor, CUSUM monitor, governor, explanation interface
K2	Explicit relations (data × control)	42010; cognitive architectures	✓ candidates never touch the prediction; control only edits the active set
K3	Principles of realisation and evolution	42010	✓ P1–P6 (above), holding at run time
K4	Fixed infrastructure, learned content	Newell; Laird (Soar)	✓ same machinery, uncalibrated, for $D = 1..50$ and dozens of external tasks
K5	Parametrised, specifiable family	42010; NAS	✓ ($d, L, \text{poles}, M_{\max}, \alpha, \text{ARL}, T_{\text{idle}}$); hash-frozen reference
K6	Non-trivial composition (ablations)	CAR-01	✓ removing self-normalisation, excitation gate, rent or warm-up breaks verified properties
K7	Topology space with a traversal rule	Cascade-Correlation; NAS	✓ sparse linear-dynamical graph whose topology changes online by test

Verdict: YES — principled online structural-learning architecture
(PRINCIPLED_ONLINE_STRUCTURAL_LEARNING_ARCHITECTURE), status
RESEARCH_SPECIFICATION_NON_CANONICAL.

Being an architecture is not the same as being competitive: on the extensive external benchmark (BENCH-02, 25 tasks, 24 models) the pre-registered decision was NOT_JUSTIFIED_EXTERNALLY (Section 7.5).

It is not: a neural-network architecture, a new learning rule, or a universal sequence model. Every primitive is prior art (Kalman, NLMS, mixture martingales, CUSUM); the claim is the *organisation*, as for Cascade-Correlation, Dyna and Horde. The model class is linear in the parameters (sparse ARX/OBF with a first-order latent state).

3. Mathematical Formulation

3.1 Predictive model and atom dictionary

The prediction is linear in the parameters over an active set $\mathcal{A}_t$ of physically meaningful structural atoms:

$$\hat{y}_t = \boldsymbol{\theta}_b^\top [\mathbf{x}_t; 1] + \sum_{a \in \mathcal{A}_t} \theta_a \varphi_a(t), \quad |\mathcal{A}_t| \leq M_{\max}$$

$$\varphi_{(i,k)}(t) = x_{t-k,i} \quad (1 \leq k \leq L), \quad \varphi_{s,p}(t) = s_{p,t-1}, \quad \varphi_{q,i}(t) = q_{i,t-1}$$

Dictionary: dL delays, $|\mathcal{P}|$ latent states (one per pole) and d drives; $p_{\text{dict}} = dL + |\mathcal{P}| + d$. At most one latent state and one drive are active (single-state limit inherited from v0.1).

3.2 Parameter learning (NLMS)

$$e_t = y_t - \hat{y}_t, \quad \boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \frac{\mu e_t \boldsymbol{\phi}_t}{\epsilon + \|\boldsymbol{\phi}_t\|^2}, \quad \mu = 0.1 \Rightarrow \mathcal{M} \approx \frac{\mu}{2-\mu} \approx 5.3\%$$

ESTABLISHED: mean-square stability for $0 < \mu < 2$ and misadjustment $\approx \mu/(2-\mu)$ under approximately white regressors (Slock 1993; Haykin).

3.3 Latent pathway in innovations form

For a hidden state $z_t = az_{t-1} + bx_{t,i} + \nu_t$ observed as $y_t = \mathbf{w}^\top \mathbf{x}_t + cz_t + \varepsilon_t$, the steady-state Kalman predictor is linear in first-order filters of the innovation and of the input, with pole $p = a(1 - K)$. LEBRE implements it with a pole bank:

$$u_t = e_t + \sum_{a \in \mathcal{A}_t^{\text{lat}}} \theta_a \varphi_a(t), \quad s_{p,t} = p s_{p,t-1} + (1-p) \frac{u_t}{\hat{\sigma}_t}, \quad q_{i,t} = p^* q_{i,t-1} + (1-p^*) x_{t,i}$$

$$p \in \mathcal{P} = \{0, 0.5, 0.8, 0.95\} \quad (\tau \approx 1, 2, 5, 20), \quad p^* = \text{pole of the active latent}$$

ESTABLISHED: equivalence of the steady-state Kalman predictor and the innovations/ARMAX form (Ljung 1999; Anderson & Moore 1979). VERIFIED: MSE on I6 at $1.17\times$ (median) and $1.64\times$ (max) of the exact optimum.

3.4 Noise scale, warm-up and excitation

$$\hat{\sigma}_t^2 = \hat{\sigma}_{t-1}^2 + \beta_t (e_t^2 - \hat{\sigma}_{t-1}^2), \quad \beta_t = \max\left(\frac{1}{t+1}, \frac{1}{n_\sigma}\right); \quad \bar{P}_t = \bar{P}_{t-1} + \beta_t^P (\|\mathbf{x}_t\|^2 - \bar{P}_{t-1})$$

$$\text{excited}_t \iff \|\mathbf{x}_t\|^2 \geq \kappa \bar{P}_t \quad (\kappa = 0.1)$$

4. Structural Lifecycle and Statistical Decision Layer

Four states: DORMANT (in the dictionary) → PROVISIONAL (under formal test, not touching the prediction) → ACTIVE (in the prediction) → EVICTED (back to the dictionary). Decisions every 10 steps; tests and evidence updated every 2 steps and only under excitation.

4.1 Screening (DORMANT → PROVISIONAL)

A rotating queue probes 2 delay candidates per step and keeps an EMA score of $e_t x_{t-k,i}$. The best $H = 4$ open test episodes with *fresh* statistics: screening chooses *what* to test and never reuses its data in the test (temporal sample splitting). Latent states are always under test while no latent is active; drives are tested only after the latent is promoted.

4.2 Promotion (PROVISIONAL → ACTIVE)

$$S_t = \sum_{j \leq t} e_j \varphi_c(j), \quad Q_t = \sum_{j \leq t} (e_j \varphi_c(j))^2, \quad \tau = \frac{\rho_{\text{rel}}}{\hat{\sigma}_{t_0}^2 m_\varphi}$$

$$\log M_t = \frac{\tau S_t^2}{2(1 + \tau Q_t)} - \frac{1}{2} \log(1 + \tau Q_t) \geq \log \frac{p_{\text{dict}}}{\alpha} \Rightarrow \text{promote}, \quad \theta_0 = \frac{S_t}{Q_t / \hat{\sigma}_t^2 + m_\varphi / \rho_{\text{rel}}}$$

ESTABLISHED: with conditionally symmetric increments, $\exp(\lambda S_t - \lambda^2 Q_t / 2)$ is a supermartingale for every λ (de la Peña 1999, Lemma 6.1; Howard et al. 2020, Lemma 3(d)); the Gaussian mixture over λ is also a supermartingale, and by Ville's inequality $P(\exists t : M_t \geq p/\alpha) \leq \alpha/p$. τ is fixed when the episode opens (predictable). VERIFIED: 0 false promotions in 400 null streams. Limit: in the adaptive loop validity is approximate; the guarantee is per episode.

4.3 Eviction (ACTIVE → EVICTED)

$$\ell_t(a) = \frac{c_a(2e_t + c_a)}{2\hat{\sigma}_t^2}, \quad c_a = \theta_a \varphi_a(t); \quad R_t(a) = \max\left(0, R_{t-1}(a) - \ell_t(a) + r \min(1, \varphi_a^2/v_a)\right)$$

$$R_t(a) > h = \log(\text{ARL}) \Rightarrow \text{evict}, \quad r = h/T_{\text{idle}}$$

ESTABLISHED: ℓ_t is exactly the Gaussian log-likelihood ratio between the "with" and "without" predictives; CUSUM has ARL to false alarm $\geq e^h$ (Lorden 1971) and is minimax optimal (Moustakides 1986) in the simple case. In LEBRE the hypotheses are composite, so optimality does not transfer. VERIFIED: 4 false evictions in 1,155,800 monitored steps.

4.4 Coupling and budget rules

- at most one eviction and one promotion per decision;
- full budget: replace only victims with $R \geq h/2$; a drive never displaces its own latent;
- evicting the latent evicts the coupled drive and closes drive tests;
- no episode opens before warm-up ($t < 20$).

4.5 Explanation interface

At every instant the model emits: base weights; each active atom with its physical meaning ("x1 delayed 6 steps", "latent state $\tau \approx 5$ driven by x0"), coefficient, accumulated evidence (nats), opposing CUSUM and activation time; and the complete lifecycle event log.

5. Resource Governance and Stream-Time Invariance

Resources are governed at three levels: (i) structural budget $M_{\max} = 4$ atoms; (ii) parsimony rent, which requires each atom to yield at least $r = h/T_{\text{idle}} \approx 0.0174$ nats per excited step; (iii) fixed cadences for screening (2/step), testing (1 per 2 steps), evidence (1 per 2 steps) and decisions (1 per 10 steps).

5.1 Stream-time invariance (ARB10 lesson)

In v0.2, decimating arbitration with a per-event α doubled the evidence timescale. In v0.3.2 no statistic is a per-event EMA: test sums, CUSUM and σ^2 evolve per sample, and when evidence is subsampled the rent is scaled by `evidence_every` to keep the per-step rate. Changing a cadence only changes the waiting time, bounded by the cadence itself.

5.2 Scale equivariance

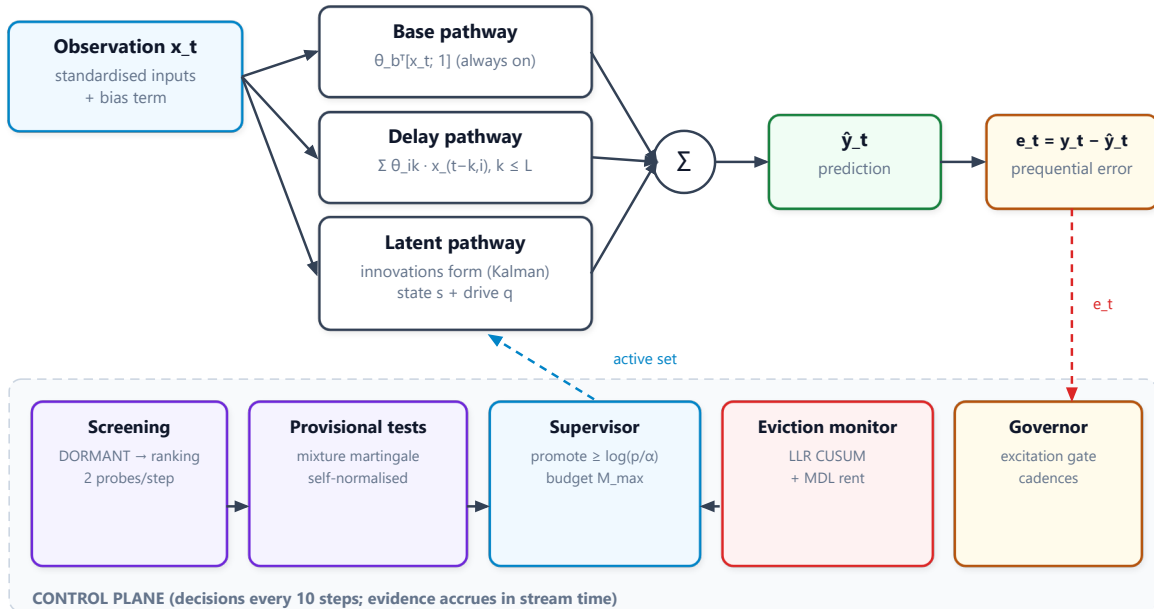
All features are dimensionless: inputs standardised by the environment, normalised innovation $u/\hat{\sigma}$, filters of standardised inputs, and a unit-information prior relative to $\hat{\sigma}$. VERIFIED: NMSE 0.2042 / 0.2042 / 0.2042 with $y \times 10^{-3} / \times 1 / \times 10^3$.

6. Architectural Diagrams (D1–D5)

D1

D1: LEBRE v0.3.2 architecture — data plane and control plane

Sparse linear-dynamical causal prediction governed by stream-time statistical tests



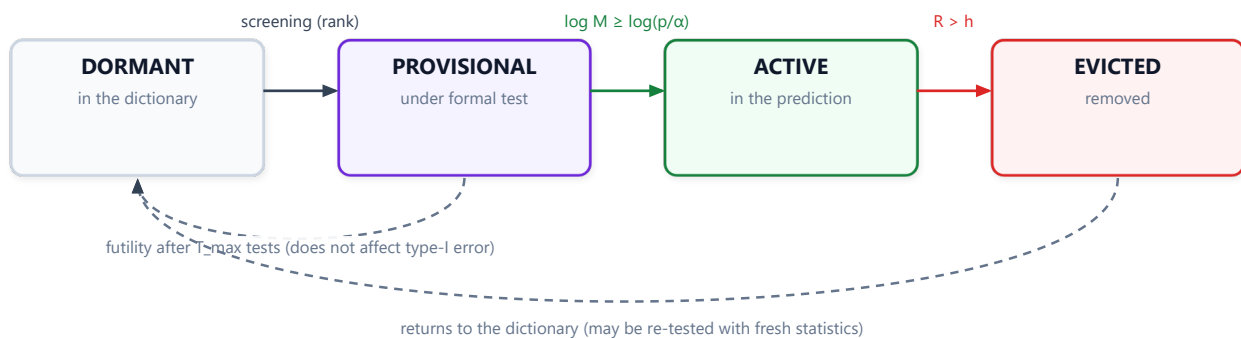
Candidates never enter the prediction before promotion (shadow probation).

Data plane (prediction) separated from the control plane (lifecycle).

D2

D2: Structural lifecycle and its transition rules

Every transition is a statistical decision with a controlled error rate



- Promotion: the mixture martingale crosses $\log(p/\alpha)$; the initial coefficient is the posterior mean.
- Eviction: the CUSUM of the 'without vs. with' log-likelihood ratio crosses $h = \log(\text{ARL})$.
- Replacement (full budget): only if the victim already has $R \geq h/2$.
- Coupling: evicting the latent state evicts its drive atom.
- Silence (inputs < 10% of long-run power): no evidence is accumulated.

Lifecycle state machine and coupling rules.

D3

D3: Provisional test episode (promotion)

Self-normalised mixture martingale: valid at any time (Ville's inequality)



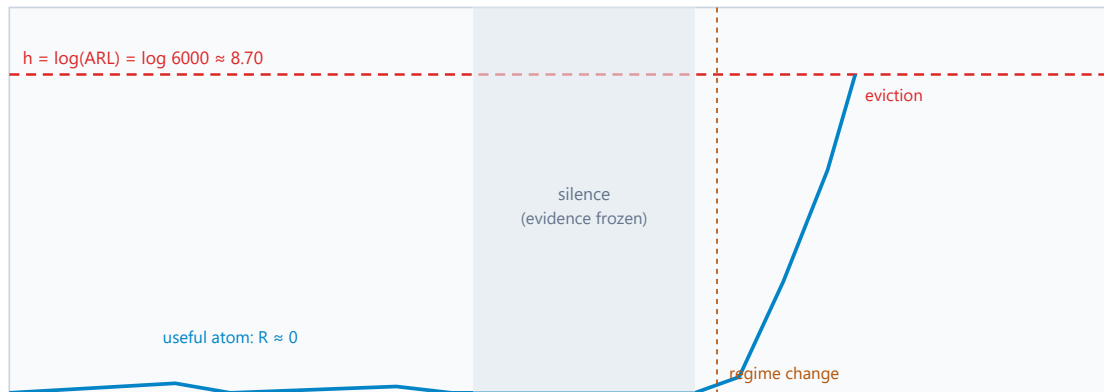
$$\log M_t = \tau S^2 / (2(1 + \tau Q)) - \frac{1}{2} \log(1 + \tau Q), \quad S = \sum e \cdot \varphi, \quad Q = \sum (e \cdot \varphi)^2$$

Under H_0 with conditionally symmetric noise: $P(\exists t: M_t \geq p/\alpha) \leq \alpha/p$ per candidate and episode.*Illustrative test episode (not measured data).*

D4

D4: CUSUM eviction with parsimony rent

A useful atom accrues evidence in its favour; silence yields no evidence; a regime change triggers eviction



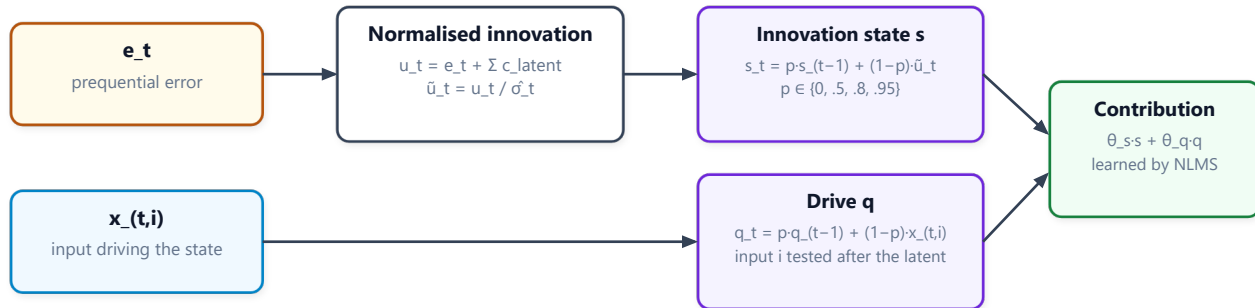
$$R_t = \max(0, R_{t-1} - \ell_t + r \cdot \min(1, \varphi^2/v_{\text{ref}})), \quad \ell_t = c(2e + c)/(2\sigma^2)$$

Without rent: ARL to false eviction $\geq e^h$ (Lorden). Rent $r = h/T_{\text{idle}}$ removes idle atoms in $\sim T_{\text{idle}}$ steps.*Illustrative CUSUM with rent and freezing in silence.*

D5

D5: Latent pathway in innovations form

Steady-state Kalman predictor of a first-order hidden state, linear in the parameters



$$z_t = a \cdot z_{t-1} + b \cdot x_{t,i} + v_t \Rightarrow \hat{y}_t \supset \theta_s \cdot s_t + \theta_q \cdot q_t \text{ with pole } p \approx a(1-K)$$

- Replaces the v0.2 RTRL recurrent unit (~20 FP) with ~9 FP of first-order filters.
- The pole bank {0, 0.5, 0.8, 0.95} covers $\tau \approx 1, 2, 5$ and 20 steps; one pole is chosen by test.
- Verified: MSE at $1.17\times$ (median) of the exact Kalman predictor on task I6.

Latent pathway: steady-state Kalman predictor via first-order filters.

7. Empirical Validation

All evaluations were pre-registered with a hash before execution, use seeds never used in development, and keep v0.3.2 frozen and uncalibrated. External baselines were calibrated per task.

7.1 Verified properties

#	Property	Result	
T1	Causality: perturbing y_t and the future leaves $\hat{y}_{\leq t}$ unchanged	max diff 0.0	✓
T2	Delay alignment ($k = 1, 7, 32$)	30/30	✓
T3	False promotions per stream (gauss / t_3 / hetero), bound 0.052	0.000 / 0.000 / 0.000	✓
T4	False evictions of the true atom	4 / 1,155,800	✓
T5	Latent MSE / optimal Kalman (median; max)	1.17; 1.64	⚠
T6	Latent retained through silence; events during silence	30/30; 0.00	✓
T7	NMSE with $y \times 10^{-3}$ / $\times 1$ / $\times 10^3$	0.2042 / 0.2042 / 0.2042	✓
T8	FP/step on the most expensive task (l4)	104.3	⚠

7.2 Internal benchmark I1–I14 (confirmatory; seeds 2146..2175)

Arm	NMSE	FP/step	Exact structure
V032	0.2186	96.9	88.6%
V031	0.2233	97.7	85.8%
V02_A0	0.3091	111.2	75.2%
CTRL_ARX	0.3368	44.0	—

H	Contrast	Δ	95% bound	Favourable	p (Holm)
H1'	V032-V02_A0 NMSE	-0.0905	-0.0822	30/30	1.7e-17
H3'	V032-CTRL_ARX NMSE	-0.1183	-0.1125	30/30	4.5e-25
H4'	V032-V02_A0 struct	+0.1348	+0.0989	25/30	2.9e-07

7.3 BENCH-01 with permuted columns (seeds 161..190; 20 models)

The true dependencies of tasks A2–A4 and H1 lie in column 0; permutation removes the artefact that favoured models looking only at that column.

Model	Mean rank	Mean NMSE	FLOPs	Pareto
LEBRE_V03	3.1	0.268	211	5
LEBRE_V031	3.9	0.291	209	0
LEBRE_V032	4.0	0.293	209	2
S3_RSONN	6.9	0.713	631	1
CTRL_ARX_NLMS	7.9	0.559	158	5
B1_RZA_LMS	7.9	0.857	241	1
B5_ONLINE_ESN	9.4	0.920	1927	0
S4_ACESN	9.6	0.896	4573	0

7.4 Never-used real data (UCI; semi-held-out)

Dataset	Persistence	ARX-NLMS	LEBRE v0.3.2	Best baseline
X1 Appliances_Energy	0.520	0.821	0.698	0.611
X2 Beijing_PM25	0.066	0.085	0.121	0.225
X3 Metro_Traffic	0.170	1.564	5.395	0.977

On the three hourly real datasets trivial persistence beats every model; on X3 a single erroneous dataset record (rain = 9831 mm/h) dominates the error of every linear model, LEBRE included.

7.5 BENCH-02 — extensive external benchmark (4 literature tracks)

Tracks: A — online time-series forecasting (ETTh1/h2/m1/m2, Exchange, Jena Weather, ECL, Traffic; FSNet/OneNet protocol at $h = 1$); B — streaming regression with drift (Friedman LEA/GRA/GSG, Planes2D, Mv, Bikes, WaterFlow, TrumpApproval; River); C — nonlinear identification (Wiener–Hammerstein, Cascaded Tanks, EMPS; one-step prediction on the official test split); D — sparse FIR system ($K = 1/4/8$, white and coloured input, abrupt switch). Baselines: the 15 calibrated BENCH-01B models, classical adaptive filters (NLMS, RLS, RZA-LMS, IPNLMS), streaming learners (River: SGD, HATR, ARF-Reg) and controls (persistence, seasonal naive, ARX-NLMS).

Pre-registered decision: NOT_JUSTIFIED_EXTERNALLY — global position 6 (mean rank 8.44); rank 1 on 2/25; top-3 on 4/25; skill > 0 vs persistence on 14/25; E3 = 0.16.

Model	Rank overall	Track A	Track B	Track C	Track D	#1	Top-3	Skill med.
IPNLMS	3.40	3.6	4.1	5.7	1.0	6	16	+0.024
CTRL_ARX_NLMS	7.20	2.9	7.5	4.3	14.0	3	8	+0.142
RIVER_SGD_LINREG	7.80	10.4	9.0	3.7	4.8	0	8	+0.013
B2_CCN	8.12	9.2	10.1	5.0	5.5	0	4	-1.453
RIVER_HATR	8.24	7.2	5.2	8.7	13.3	2	5	-0.065
LEBRE_V032	8.44	7.4	6.9	6.7	12.8	2	4	+0.506
RIVER_ARF	9.36	5.5	5.1	9.0	20.3	3	5	-0.045
CTRL_SEASONAL_NAIVE	9.57	9.6	—	—	—	0	1	-10.205
S3_RSONN	9.88	9.8	5.9	11.3	14.7	1	3	-0.094
CTRL_PERSISTENCE	10.20	1.6	13.2	2.0	21.7	8	12	+0.000

LEBRE v0.3.2 per task

Track	Task	NMSE	Skill	Rank	Best other	NMSE
A	A_ECL	0.0861	+0.112	1/23	CTRL_PERSISTENCE	0.0970
A	A_ETTh1	0.0396	-1.048	6/23	CTRL_ARX_NLMS	0.0190
A	A_ETTh2	0.1356	-8.908	11/23	CTRL_PERSISTENCE	0.0137
A	A_ETTm1	0.0273	-5.186	8/23	CTRL_PERSISTENCE	0.0044
A	A_ETTm2	0.0481	-33.049	8/23	CTRL_ARX_NLMS	0.0012
A	A_Exchange	0.0250	-24.109	15/22	CTRL_PERSISTENCE	0.0010
A	A_JenaWeather	0.0041	-3.295	5/23	CTRL_ARX_NLMS	0.0009
A	A_Traffic	0.2174	-0.871	5/23	CTRL_PERSISTENCE	0.1162
B	B_Bikes	0.6773	+0.506	4/22	S3_RSONN	0.6678
B	B_FriedmanDrift_GRA	0.3024	+0.848	2/22	RIVER_HATR	0.2803
B	B_FriedmanDrift_GSG	0.3103	+0.844	2/22	RIVER_HATR	0.2856
B	B_FriedmanDrift_LEA	0.3626	+0.819	4/22	RIVER_ARF	0.2436
B	B_Mv	0.3756	+0.812	5/22	RIVER_ARF	0.0628
B	B_Planes2D	0.3080	+0.846	11/22	RIVER_ARF	0.1386
B	B_TrumpApproval	0.4313	-8.445	9/22	CTRL_PERSISTENCE	0.0457
B	B_WaterFlow	4.9816	-82.138	18/22	CTRL_PERSISTENCE	0.0599
C	C_CascadedTanks	0.0076	-1.778	11/22	CTRL_PERSISTENCE	0.0027
C	C_EMPS	0.0359	-29570.687	8/22	CTRL_PERSISTENCE	0.0000
C	C_WienerHammerstein	0.0032	+0.822	1/22	B2_CCN	0.0177
D	D_SparseFIR_K1_colored	0.0669	+0.846	14/23	IPNLMS	0.0218
D	D_SparseFIR_K1_white	0.0579	+0.971	14/23	IPNLMS	0.0207
D	D_SparseFIR_K4_colored	0.0553	+0.876	12/23	IPNLMS	0.0187
D	D_SparseFIR_K4_white	0.0550	+0.973	13/23	IPNLMS	0.0165
D	D_SparseFIR_K8_colored	0.0536	+0.854	11/23	IPNLMS	0.0202
D	D_SparseFIR_K8_white	0.0661	+0.969	13/23	IPNLMS	0.0211

8. Computational Accounting and Embedded Envelope

Component	FP/step	Scaling
Prediction + NLMS (base)	$\approx 6d + 13$	$O(d)$
Per active atom (prediction, NLMS)	6	$O(A)$
CUSUM evidence per atom (1 per 2 steps)	≈ 4.5	$O(A)$
Latent bank (3 poles + normalisation)	≈ 11	$O(P)$
Screening (2 probes)	8	$O(1)$
Provisional tests (1 per 2 steps)	≈ 2 per candidate	$O(H + P)$
Decisions (1 per 10 steps)	$\approx 1-7$	$O(H + P)$

Not counted as FP: floating-point comparisons in the top-k dictionary selection ($\approx p_dict$ comparisons per refill) and log calls (≈ 0.001 per step). On I1–I14 ($d = 5$) the mean is 96.9 FP/step, ≤ 100 , but the most expensive task (I4) reaches 104.3. With $d = 10..50$ the cost is $O(d)$ ($\approx 110-370$ FP) and memory (history of $(L+1) \cdot d$ samples + screening scores) exceeds 1 KB. fp16 storage is an accounting convention not yet numerically validated.

9. Scope Limits and Forbidden Claims

Limit	Description
Model class	linear in the parameters; one first-order latent state; delays $\leq L = 32$
Hourly real data	trivial persistence wins on X1–X3; target-autoregressive atoms are missing
Input outliers	unprotected; one erroneous record dominates the error (X3)
High-d search	rotating screening takes $\approx dL/2$ steps per cycle; discovery is slow for $d \geq 20$
Guarantees	exact only under idealised hypotheses; approximate in the adaptive loop; CUSUM optimality not transferred
Embedded envelope	≤ 100 FP and ≤ 1 KB only for small d ; hardware not validated
External competitiveness (BENCH-02)	6th of 24 models; pre-registered decision NOT_JUSTIFIED_EXTERNALLY; IPNLMS (2002) wins all 6 sparse-FIR conditions and ARX/persistence win hourly forecasting
Adaptation step	fixed μ with no per-coefficient proportional gain; slow adaptation on near-unit-root series (hypothesis)
Missing comparators	DMA, alpha-investing, ARX with RIC and deep online forecasters (FSNet, OneNet)

Forbidden claims

- novelty or scientific priority (NOVELTY_CLAIM_READY = NO)
- state of the art in time-series forecasting
- energy efficiency or MCU feasibility from FP counts
- exact error guarantees in the adaptive loop
- replacement of canonical LEBRE v0.1
- global validation or opening of M3

10. Parameter Traceability Matrix

Parameter	Value	Origin / role	Status
d	—	task dimension	—
L	32	inherited from v0.2	DESIGN
μ	0.1	misadjustment $\mu/(2-\mu) \approx 5.3\%$	DERIVED
n_σ	200	noise-scale memory	DESIGN
α	0.05	family-wise error per episode	DESIGN
$\log(p_{\text{dict}}/\alpha)$	8.2 (d=5) ... 10.4 (d=50)	Ville + Bonferroni	DERIVED
ARL, h	6000, 8.70	$h = \log \text{ARL}$ (Lorden)	DERIVED
T_{idle}, r	500, 0.0174	$r = h/T_{\text{idle}}$	DERIVED / DESIGN
ρ_{rel}	1.0	unit-information prior	DERIVED
$\mathcal{P}$	{0, 0.5, 0.8, 0.95}	$\tau \approx 1, 2, 5, 20$	DESIGN
M_{max}	4	structural budget	DESIGN
B_{probe}, H	2, 4	screening / concurrent tests	DESIGN
screen_w	0.2	only chooses what to test	DESIGN
decide / test / evidence	10 / 2 / 2	cadences (bounded wait)	DESIGN
T_{max}	200 tests	futility (type-I unaffected)	DESIGN
κ, n_P	0.1, 1000	excitation gate	DESIGN
n_{warm}	20	warm-up (F6)	DESIGN
$R \geq h/2$	—	replacement rule	DESIGN
σ^2 floor	$10^{-10} \cdot \text{peak}$	numerical robustness	DESIGN

No parameter was tuned on the validation benchmarks. The v0.3 values were fixed in 4 DEV iterations (seeds 3301..3310); fixes F1–F6 came from property tests and one diagnosed failure, and were evaluated on fresh seeds.

11. Architectural Decision Records and Final Decision

ADR	Decision	Consequence
ADR-301	Every structural decision is a test with controlled error	replaces gain EMAs and hand-tuned thresholds
ADR-302	Stream-time evidence	decision decimation does not distort timescales
ADR-303	Innovations-form latent + drive	replaces RTRL; ~9 FP; explainable
ADR-304	Self-normalised evidence and scale equivariance	validity under heavy tails and heteroscedasticity
ADR-305	Silence is not evidence	persistent-excitation gate
ADR-306	Structure pays rent	excitation-weighted MDL rent
ADR-307	Screening chooses, the test decides	temporal sample splitting
ADR-308	Single latent state and budget $M_{\max}$	scope limit inherited from v0.1
ADR-309	Sample-mean warm-up estimators	fixes the $\tau \approx 10^{13}$ blocking of v0.3.1

Final decision

ARCHITECTURE_CLASSIFICATION = PRINCIPLED_ONLINE_STRUCTURAL_LEARNING_ARCHITECTURE
STATUS = RESEARCH_SPECIFICATION_NON_CANONICAL
INTERNAL_EVIDENCE = CONFIRMED (H1'-H4', Holm)
EXTERNAL_EVIDENCE = NOT_JUSTIFIED_EXTERNALLY
CANONICAL_REFERENCE = LEBRE v0.1 (FROZEN_WITH_SCOPE_LIMITS)
NOVELTY_CLAIM_READY = NO · M3 = UNOPENED

Key references

- Anderson & Moore (1979) *Optimal Filtering*
- Chen, Gu & Hero (2009) Sparse LMS for system identification, ICASSP
- Benesty & Gay (2002) An improved PNLMS algorithm, ICASSP
- de la Peña (1999) *Ann. Probab.* 27(1)
- Elsken, Metzen & Hutter (2019) Neural Architecture Search: A Survey, *JMLR* 20
- Fahlman & Lebiere (1990) The Cascade-Correlation Learning Architecture, NIPS 2
- Foster & George (1994) The Risk Inflation Criterion, *Ann. Statist.* 22(4)
- Gomes et al. (2018) Adaptive random forests for data stream regression, ESANN
- Howard, Ramdas, McAuliffe & Sekhon (2020) *Probability Surveys*; (2021) *Ann. Statist.* 49(2)
- Ikonomovska, Gama & Džeroski (2011) Learning model trees from evolving data streams, *DMKD*
- Ioannou & Sun (1996) *Robust Adaptive Control*

- ISO/IEC/IEEE 42010:2022 Architecture description
- Laird (2012) *The Soar Cognitive Architecture*
- Ljung (1999) *System Identification*, 2nd ed.
- Lorden (1971) *Ann. Math. Statist.* 42; Moustakides (1986) *Ann. Statist.* 14
- Page (1954) Continuous inspection schemes, *Biometrika*
- Pham et al. (2023) FSNet, ICLR; Wen et al. (2023) OneNet, NeurIPS
- Raftery, Kárný & Ettler (2010) Dynamic Model Averaging, *Technometrics* 52
- Schoukens & Noël (2017) Three benchmarks in nonlinear system identification, IFAC
- Shin, Ramdas & Rinaldo (2023) E-detectors, *NEJSDS*
- Slock (1993) *IEEE TSP* 41
- Sutton (1990) Dyna; Sutton et al. (2011) Horde, AAMAS
- Wahlberg (1991) *IEEE TAC* 36
- Zhou, Foster, Stine & Ungar (2006) Streamwise feature selection, *JMLR* 7
- Zhou et al. (2021) Informer / ETT datasets, AAAI